



ĐẠI HỌC
BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY
OF SCIENCE AND TECHNOLOGY

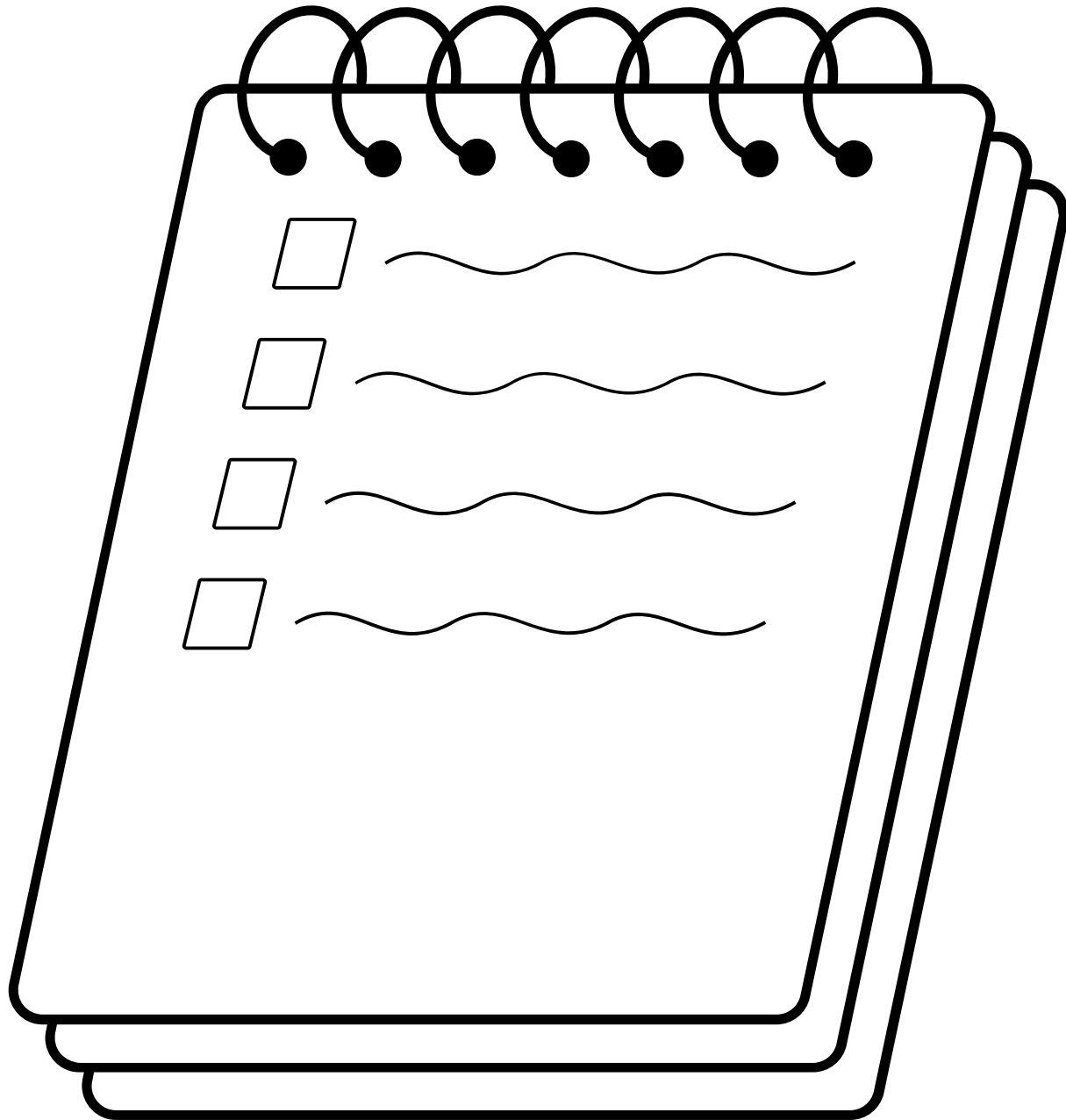
Weather Monitoring and Forecasting System

Big Data Storage and Processing

Group 13

Nguyễn Trí Thanh	20225457
Phan Trần Việt Bách	20225435
Lê Nhật Quang	20225522
Nguyễn Hoàng Sơn	20225525
Nguyễn Minh Quân	20225520

ONE LOVE. ONE FUTURE.

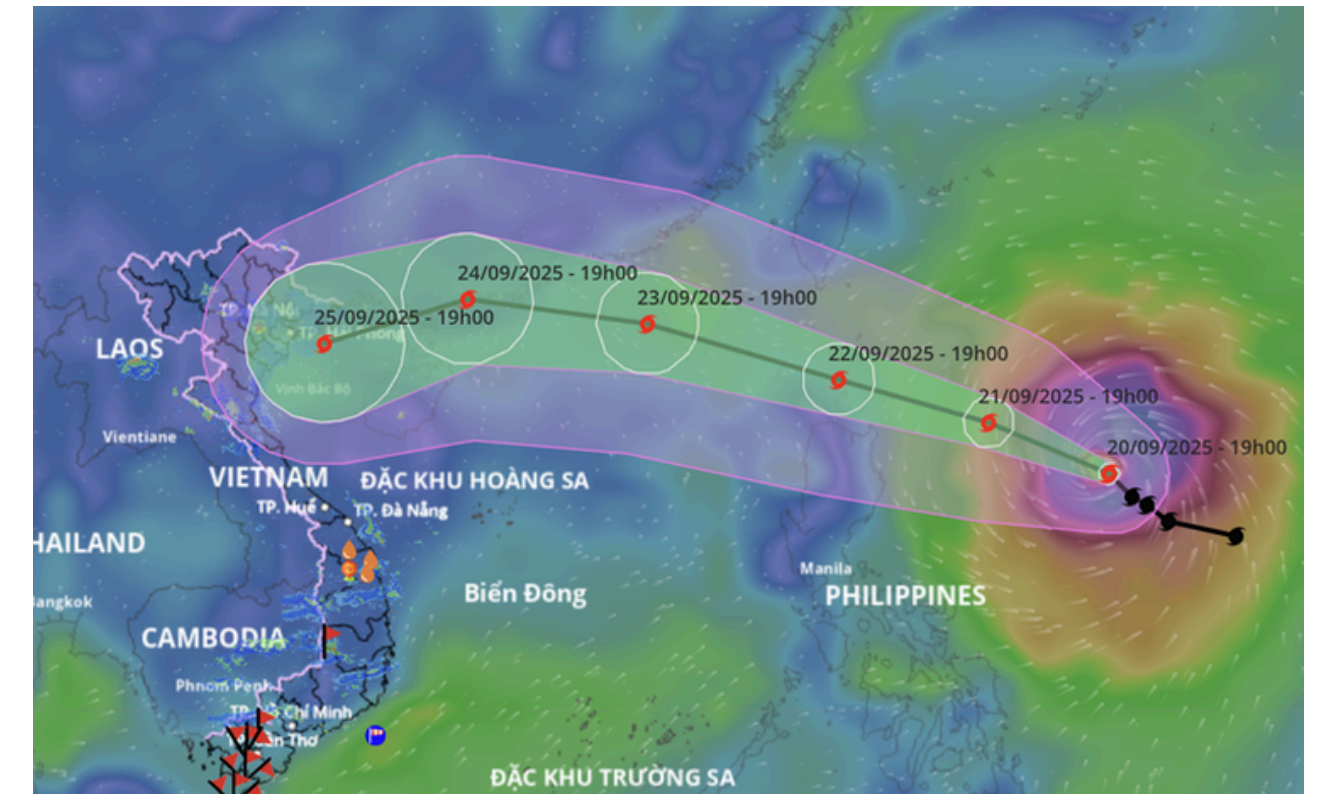


1. Introduction
2. Data Sources
3. System Architecture
4. Expected Results

1. Introduction

Motivation

- Every year, the East Sea faces numerous tropical storms and typhoons, causing severe impacts on coastal and low-lying communities.
- Timely warnings can help fishermen, local authorities, and citizens prepare effectively and minimize damage.
- Flood and inundation alerts, early water overflow and urban flooding warnings, can help communities take preventive actions before severe impacts occur.



URGENT

1. Introduction

Motivation

- Our project, **Weather Monitoring and Forecasting System** using Big Data, aims to enhance the accuracy and speed of weather prediction by analyzing vast amounts of real-time meteorological and hydrological data from satellites, sensors, and IoT devices.
- Main functions:
 - Real-time weather monitoring system
 - Forecasting future weather conditions (rainfall, sunshine, storms, ...)
 - Supporting urban flood prediction and natural disaster early warning systems.



Leveraging Multiple Data Types

Data Type	Description / Source	Common Fields	Main Contribution
Rainfall Observation (Ground Stations)	Rainfall measured by national meteorological & hydrological stations	- Station ID- Location (lat, lon, elevation)- Timestamp- Rainfall (mm)- Data status	Monitoring & Validation
Satellite / Gridded Rainfall Data	Rainfall estimated from satellites or climate models (e.g., TRMM, GPM, CHIRPS)	- Grid (lat, lon)- Time- Rainfall (mm/h)- Quality/confidence	Prediction & Spatial Monitoring
Water Level / River Discharge Data	River water level and discharge at hydrological stations	- Station ID- Location- Timestamp- Water level (m), discharge (m³/s)	Monitoring & Flood Warning
Soil Moisture Data	Measured or estimated soil saturation levels	- Area/grid ID- Time- Moisture (%) by depth	Prediction (Runoff Modeling)
Topographic / Elevation Data (DEM, LiDAR)	Digital elevation model of terrain	- Grid (lat, lon)- Elevation (m)- Slope / aspect	Flood Mapping & Modeling
Land Cover / Land Use Data	Classification of land surfaces (forest, urban, agriculture, etc.)	- Grid/vector- Land type- Impervious surface ratio	Prediction & Flood Simulation
Flood Inundation Maps (Remote Sensing)	Flooded areas detected by SAR or optical satellite images	- Grid (lat, lon)- Time- Flood depth / binary- Confidence	Flood Detection & Validation
Meteorological Parameters	Temperature, pressure, humidity, wind, radiation, evaporation	- Location- Time- Variables: temp, pressure, humidity, wind, radiation, evapotranspiration	Prediction (Weather Modeling)
Urban / Social Sensor Data	IoT water sensors, CCTV, crowd reports, traffic flow	- Sensor/camera ID- Timestamp- Water depth / wetness / image	Monitoring & Early Warning

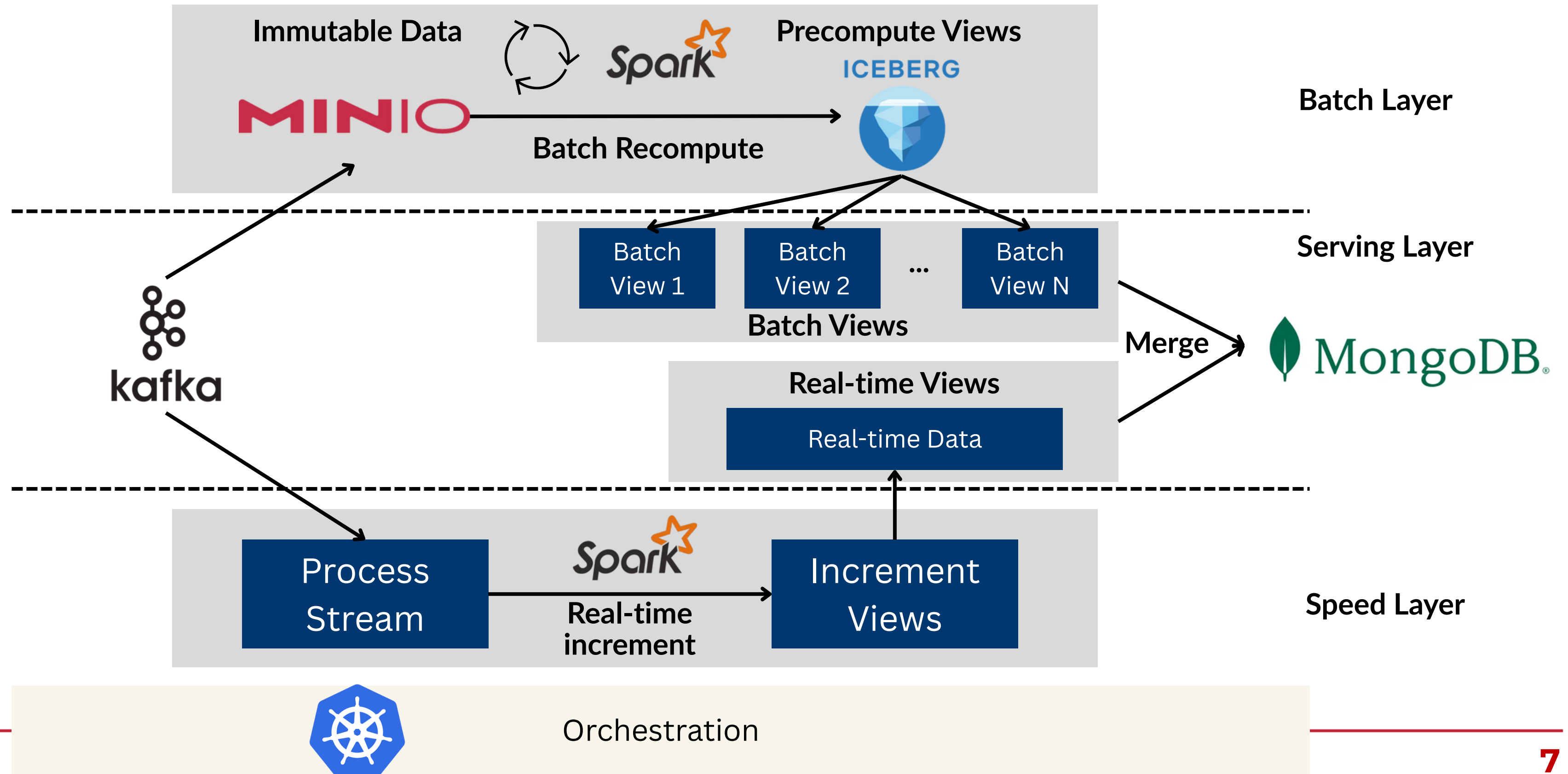
2. Data Sources

Integration of multiple data sources

- Rainfall & Weather Data: Rainfall observation, CHIRPS rainfall, GPM satellite rainfall, Open-Meteo weather & flood API, ...
- Hydrology & Flood Forecasting: GloFAS (Global Flood Awareness System), GFMS (Global Flood Monitoring System, NASA/UMD), DAHITI (Satellite Water Level Database), ...
- Remote Sensing & Flood Mapping: MODIS/VIIRS flood maps, Sentinel-1 SAR imagery, Global Flood Monitoring (Copernicus GFM), SRTM DEM (NASA/USGS), ESA WorldCover 2020 (10 m land cover)
- Meteorological & Environmental Data: NASA POWER climate & solar dataset, ERA5 reanalysis (Copernicus), Vietnam climate data 1901–2020
- Urban & Real-time Social Data: Vietnam IoT flood sensors (Smart City projects), Google Maps traffic data
- Vietnam & Regional Open Data: Open Development Vietnam portal, UN-SPIDER GIST datasets for Vietnam, VMHA / KISTERS National Flood Forecasting System

3. System Architecture

Lambda Architecture: for batch and stream processing



3. System Architecture

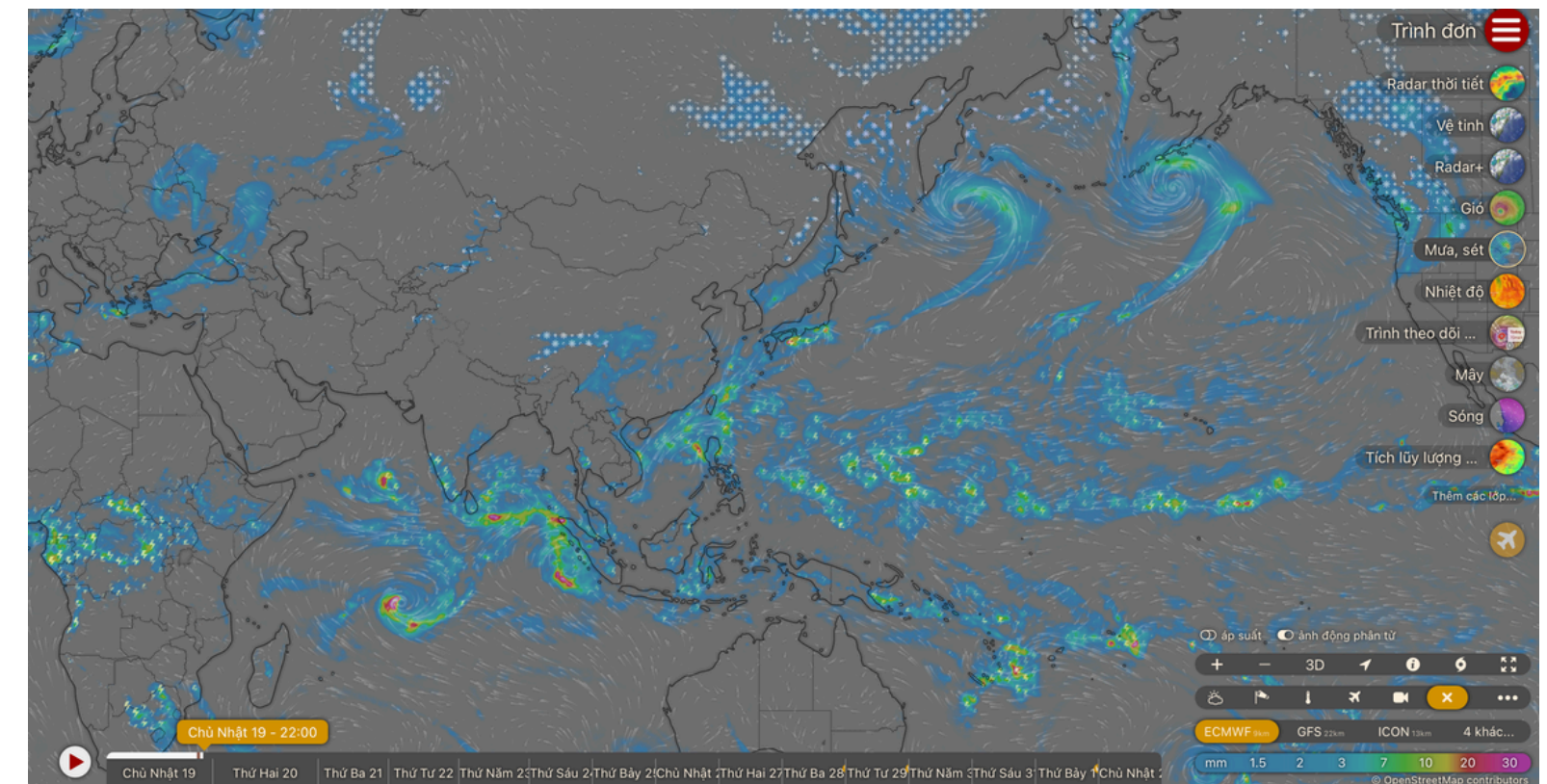
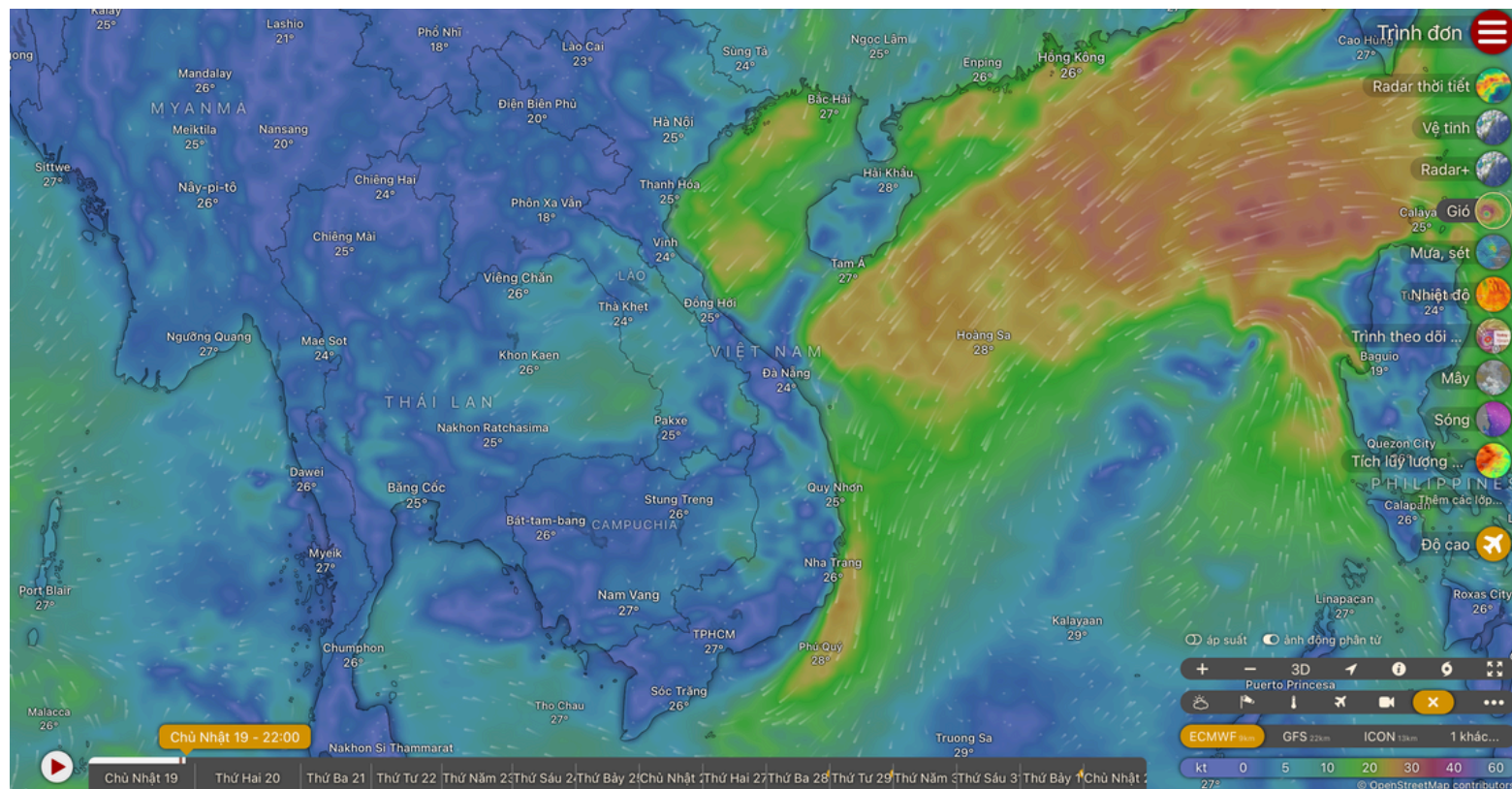
Component	Technology	Description
Message Queue	Apache Kafka	Collects and streams incoming data from various producers to downstream consumers in real time
Distributed Storage	MinIO	Stores raw input data files and holds Iceberg’s metadata and processed data files
Table format & metadata management	Apache Iceberg	Provides batch views with schema evolution, ACID transactions, and versioning via snapshots
Batch Processing	Apache Spark	Performs ETL on raw data and forms batch views
Stream Processing	Spark Structured Streaming	Consumes Kafka streams to produce real-time views with incremental updates
Serving layer	MongoDB	NoSQL database that stores merged batch and stream views, enabling fast, low-latency queries
Orchestration	Kubernetes	Manages the system’s components as pods, ensuring fault-tolerance and scalability



4. Expected Results

- A data processing pipeline leveraging both:
 - batch processing for weather forecasting
 - stream processing for real-time weather monitoring
- A visual monitoring system, integrating big data analytics and visualization tools

=> An advanced application for weather observation, forecasting, and adaptation for climate changes





HUST

THANK YOU !